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Roll No. .....

341152(41)

B. Pharmacy (First Semester) Examination,
Nov.-Dec. 2021

(PCI Scheme)

(Pharmacy Branch)

PHARMACEUTICAL ANALYSIS-I

Time Allowed : Three hours

Maximum Marks : 75

Note : Question paper contains three Sections. Read the instructions of each section carefully.

Section-A

20×1=20

Note : Attempt all questions. Each question carries 1 mark.

1. (i) API stands for :

(a) Active Pharmacy Interpretation

(b) Acute Pulmonary Interaction

(c) Active Pharmaceutical Ion

(d) Active Pharmaceutical Ingredient

(ii) Which analytical techniques is used for determine the amount of oxidizing agents?

(a) Oxidation-reduction titration

(b) Bromometry

(c) Complexometry

(d) None of these

(iii) Random error is also known as :

(a) Accidental error

(b) Indeterminate error

(c) Determinate error

(d) Both (a) and (b)

[3]

(a) Arrhenius theory

(b) Common ion effect

(c) Bronsted theory

(d) None of these

(v) Weakly acidic and weakly basic substances analyzed by which titration method :

(a) Aqueous titration

(b) Non-aqueous titration

(c) Redox titration

(d) Complexometric titration

(vi) Which method is used in water analysis :

(a) Fajan's method

(b) Mohr's method

(c) Volhard's method

(d) None of these

(vii) SI unit of conductance is :

(a) mho

(b) siemens

(c) volt

(d) None of these

(viii) Current used for measurement of conductance is

(a) A.C.

(b) D.C

(c) Any one of these

(d) None of these

(ix) Solubility of sparingly soluble salt can be determined by :

(a) Polarography

(b) Potentiometry

(c) Conductometry

(d) IR spectroscopy

(x) Potentiometry is an ..... method of analysis.

(a) Spectroscopic

(b) Analytical

(c) Electro-analytical

(d) None of these

(xi) Hydrogen electrode can be used as .....

(a) Reference

(b) Indicator

(c) Both of above

(d) None of these

(xii) There is a linear relationship exist between pH and ..... of the solution.

(a) Colour

(b) Potential

(c) Turbidity

(d) None of these

(xiii) Residual current in polarography is due to :

(a) Oxidizable impurity

(b) Reducible impurity

(c) Analyte

(d) All of these

(xiv) Diffusion current can be correlated with different conditions by :

(a) Nernst equation

(b) Bragg's equation

(c) Illkovic's equation

(d) Beer's equation

(xv) Antimoney electrode is an :

(a) Indicator electrode

(b) Secondary reference electrode

(c) Reference electrode

(d) None of these

(xvi) Which one is useful in non-aqueous solution :

(a) Leveling solvent

(b) Differentiating solvent

(c) Both (a) and (b)

(d) None

(xvii) Which of the following is an example of adsorption indicator :

(a) Eosin

(b) Phenolphthalein

(c) Methyl Red

(d) Ninhydrine

(xviii) Oxidation involve :

(a) Gain in oxygen

(b) Loss of oxygen

(c) Both (a) and (b)

(d) None of the above

(xix) Lesser the pH ..... in the acid nature.

(a) Weaker

(b) Stronger

(c) Both (a) and (b)

(d) None of these

(xx) Phenolphthalein change color ..... in

(a) Acid

(b) Alkali

(c) Water

(d) Salt

Section-B 2×10=20

(Long Answer Types Questions)

Note: Attempt any two questions. Each question carries 10 marks.

2. Discuss about gravimetric analysis along with the principle and steps involved in it.

3. Write a detail note on redox titration with principle and application of titration involving permangnametry.

4. Give detail note on Polarography its principle and instrumentation.

Section-C 7×5=35

(Short Answer Types Questions)

Note: Attempt any seven questions. Each question carries 5 marks.

5. Write down the sources of Error and its types.

6. [illegible]

7. [illegible]

8. [illegible]

9. Write a note on [illegible]

10. [illegible]

11. Write a note on [illegible]

12. Write a estimation of [illegible] by Mohr's method

13. Write a note on [illegible]